

Summary

Enterprise GenAI Engineer with 9+ years of experience designing and operating enterprise-scale AI systems, distributed platforms, and agentic AI solutions. Specialized in AI runtime architectures, Agentic AI, LangGraph workflows, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), and cloud-native platform engineering. Proven track record building production-grade AI platforms in regulated financial environments with expertise in workflow orchestration, state management, observability, and scalable AI infrastructure.

Technical Skills

Programming Languages:	Python, Java, JavaScript, TypeScript, SQL, PL/SQL, C#, HTML, CSS, Shell Scripting
Gen AI Engineering:	ACP, RAG, MCP, Ollama, HF Transformers, vLLM, Hermes Agent, Copilot, OpenCode, LangSmith, Arize
AI / Machine Learning:	LangChain, LangGraph, LangFlow, PyTorch, TensorFlow, Scikit-Learn, NLP, Deep Learning
Backend & Full-Stack Development:	Django, Django REST Framework, FastAPI, Spring Boot, ReactJS, TypeScript, Vite, MUI, AngularJS
Cloud & Platform Engineering:	AWS (EC2, S3), OpenShift, Kubernetes, Docker, Jenkins, CI/CD
Data Engineering & Databases	ElasticDB, MongoDB, PostgreSQL, Oracle, MySQL, Knowledge Graphs, SQL Optimization
Architecture & Engineering Practices:	Solution Architecture, Microservices, DevOps, Agile Scrum, System Design

Work Experience

- VP1, Enterprise GenAI Developer -United Overseas Bank**

June 2025— Present

 - Lead the end-to-end design and delivery of enterprise GenAI solutions partnering with business analysts, product stakeholders, and end users to translate business objectives into scalable, secure, and production-ready AI capabilities within a regulated banking environment.
 - Develop, and maintain enterprise-grade Java Spring Boot microservices and RESTful API with Python AI platforms to integrate LLMs, RAG pipelines, enterprise systems, and banking services through event-driven API-first architectures and real-time business process automation.
 - Architect and implement multi-agent orchestration frameworks comprising Planner, Executor, Monitoring, and Result Validation agents, leveraging LangGraph, harness engineering, and iterative feedback loops to enable autonomous task decomposition, workflow execution, quality evaluation, self-correction, and continuous refinement of complex business processes.
 - Drive production readiness through CI/CD automation, observability, structured logging, governance controls, resilience patterns, and operational best practices to ensure reliability, performance, and security at enterprise scale.
- Data Scientist - Scientific Infra & Private Assets (SIPA)**

November 2023— May 2025

 - Led the architecture, development, and operationalization of an agentic AI transaction-intelligence platform, using LangGraph to orchestrate multi-step RAG workflows for financial research, due diligence, and contextual insight generation from internal and external data.
 - Designed scalable LLM-based entity-intelligence and data pipelines for entity extraction, relationship analysis, knowledge-graph generation, and the storage and retrieval of structured and unstructured financial information.
 - Built the end-to-end full-stack dashboard application, enabling analysts to investigate transactions, review AI-generated outputs, visualize entity relationships, and manage financial screening workflows through a centralized interface.
 - Owned application deployment and production operationalization on AWS, including EC2-hosted services and S3-based storage while managing containerized components, system integration, release delivery, and application troubleshooting.
- Senior Software Consultant - znt Singapore Pte. Ltd**

February 2021— October 2023

 - Designed and delivered industrial automation and distributed software solutions by analyzing production workflows, integrating heterogeneous systems, and driving process optimization across large-scale manufacturing environments.
 - Diagnosed and resolved complex cross-platform issues through architecture analysis, database optimization, logging, and code-level troubleshooting, improving system reliability, performance, and operational resilience.
- Senior Application Engineer - ATS Applied Tech Systems Pte Ltd**

September 2019 — February 2021

 - Developed a Remote Operating System enabling seamless machine-to-command center communication, with a robust cross-platform GUI using Ignition SCADA, Java, and Python.
 - Designed SCADA and MES systems to simulate automated production lines, implementing workflows with sequential function charts and generating real-time dashboards and reports.
- Instrumentation and Control Engineer - Winsys Technology Pte Ltd**

August 2017 — September 2019

 - Developed an uninterrupted monitoring system with the hot standby environment and the configuration of historian and server redundancy.
 - Designed the system architecture of the SCADA monitoring system and configured server virtualization and database redundancy.

Education

National University of Singapore

Institute of System Science - Master of Technology (Intelligent Systems)

Jan 2021 — Jan 2023

CGPA: 4.35-/5.00

Nanyang Technological University

School of Mechanical and Aerospace Engineering - B.Eng (Mechanical Engineering), Mechatronics Specialization

August 2013 — July 2017

CGPA: 4.30-/5.00

Project Experience

Build Your Own Agent (BYOA)

- United Overseas Bank

Apr 2026 - Present

- Architected and developed an agentic service layer on top of the multi-tenant Build-Your-Own-Bot platform to enable tenant chatbots to securely discover and invoke configurable AI agents during user conversations.
- Built a tenant-specific management interface for defining and maintaining defined agents, skills, references and role-based access policies with authorized MCP servers, including NumExpr math agent, document Assistant to enable specialized tools and enterprise knowledge.
- Integrated tenant-defined configurations with LangChain Deep Agents and applied custom middleware to dynamically load tenant-defined skills, intercept and log tool calls and govern behavior throughout multi-step agent execution loops.

Build-Your-Own-Bot (BYOB)

- United Overseas Bank

Jun 2025 - Present

- Architected and developed a multi-tenant enterprise GenAI platform enabling business units to create, configure, and manage domain-specific AI assistants within governed workspaces, supported by role-based access controls and proprietary enterprise knowledge sources.
- Designed and implemented an end-to-end RAG architecture covering document ingestion, chunking, embedding generation, and metadata-driven retrieval through ElasticDB vector search, enabling configurable knowledge scopes across document groups and categories.
- Engineered multiple Python microservices using FastAPI and Uvicorn to provide AI, retrieval, and agentic capabilities to the core Java chatbot service through well-defined service interfaces, creating a modular architecture that separated AI workloads from enterprise application logic.
- Integrated privately hosted open-source LLMs served through vLLM with LangChain, LangGraph, and MCP-enabled tools to support dynamic tool orchestration, stateful multi-step workflows, and automated task execution.
- Established a cloud-native deployment and DevOps framework using OpenShift, Docker, Jenkins CI/CD, ReactJS, Java enterprise services, and Python microservices, enabling secure, independently maintainable, and production-ready GenAI platform operations.

Data Team Dashboard

- Scientific Infra & Private Assets (SIPA)

Nov 2023— May 2025

- Architected and developed an enterprise full-stack platform for transaction intelligence and workflow automation using Django, ReactJS (TypeScript), MongoDB Atlas, PostgreSQL, Docker, Gunicorn, and Nginx.
- Built an Agentic AI-powered financial intelligence workflow using LangChain and LangGraph to orchestrate multi-step information gathering, analysis, and report generation from structured and unstructured data sources.
- Developed a Retrieval-Augmented Generation (RAG) pipeline leveraging DuckDuckGo Search as an external retrieval tool to enrich internal transaction records with real-time web intelligence, enabling contextualized financial summarization and improving entity discovery accuracy through combined analysis of structured and unstructured data.
- Built an LLM-driven entity intelligence platform that performed prompt-based NER and relationship extraction to identify organizations, individuals, subsidiaries, and SPVs, automatically generating knowledge graphs for due diligence and ownership analysis while leveraging optimized MongoDB schemas and aggregation pipelines for scalable data management.

Central Knowledge Source

- Capstone project, M.Tech (Intelligent Systems)

January 2022— Oct 2022

- Developed a centralized enterprise search application using Sentence-BERT embeddings and TF-IDF, enabling users to retrieve relevant information from shared-drive documents and internal Confluence webpages.
- Engineered a full-stack Django and ReactJS application with automated ETL pipelines to extract and structure content from PowerPoint, Word, PDF, and web sources, integrating ML-powered semantic search through Django backend services.
- Designed an extensible OracleDB relational data model to organize extracted content and relationships across heterogeneous sources, deploying the application on OpenShift with Git, Bitbucket, and CI/CD workflows.

Conversational Natural Language Query of Relational and Non-relational Databases

- Practice Module, Practical Language Processing of M.Tech (Intelligent Systems)

August 2022— November 2022

- Developed a full stack Django application (Django+React+Vite) as a chatbot solution to translate natural language into SQL or GQL.
- Performed intent classification and Q&A to extract the utterance and intention from users by using Sentence Transformer and TFIDF Vectorizer.
- Analyzed the entities and POS tags of the user query and generated a rule-based algorithm to perform natural language generation.